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Summative Course Assessment

Graded Assignment: Course Summative Assessment

Your grade: 80%

Your score: 80% · Your highest

Course Summative Assessment

Instructions

Review Learning Objectives

1. What is the primary characteristic of cross-sectional data?
- ☐ It combines both cross-sectional and longitudinal data.
- ☒ It provides a snapshot at a single point in time.
- ☐ It tracks the same subjects over multiple time points.
- ☐ It is collected at regular intervals.

Correct
Due: Dec 22, 11:59 PM CST
Attempts: 0 left (3 attempts every 8 hours)
Submitted: Nov 17, 1:35 AM CST
Try again

2. Which data type tracks the same subjects over multiple time points?
- ☒ Longitudinal Data
- ☐ Panel Data
- ☐ Cross-sectional Data
- ☐ Time-series Data

Correct
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3. If you have a normal distribution and you want to estimate the mean and variance using maximum likelihood inference, which of the following is the correct likelihood function?
- ☐ The sum of the absolute differences between each data point and the mean.
- ☐ The ratio of the sum of the data points to the number of data points.
- ☐ The sum of the squared differences between each data point and the mean.
- ☒ The product of the probabilities of each data point given the mean and variance.

Correct
Correct: The likelihood function for a normal distribution is the product of the probabilities of each data point given the mean and variance.

4. Which type of data could be described as "higher dimensional"?
- ☐ Longitudinal data
- ☐ Categorical Data
- ☒ Panel Data
- ☐ Binary Data

Incorrect

5. What does a high posterior probability indicate in Bayesian inference?
- ☐ The prior beliefs are very accurate.
- ☐ All observed data are outliers.
- ☒ The observed data is consistent with the prior beliefs.
- ☐ The likelihood of the data is very high.

Correct
Correct: A high posterior probability indicates that the observed data is consistent with the prior beliefs.

6. When is it appropriate to use a data transformation method over ordinary least squares regression?
- ☐ When the number of predictors is greater than the number of observations.
- ☐ When the data perfectly fits the assumptions of linear regression.
- ☐ When the dependent variable is binary.
- ☒ When the data contains outliers or are heteroskedastic.

Correct

7. In LDA, what happens if the assumption of equal covariance matrices across different classes is violated?
- ☐ The method becomes equivalent to PCA.
- ☐ LDA will always fail to classify any observations correctly.
- ☒ The performance of LDA may degrade, potentially leading to less accurate classifications.
- ☐ The number of discriminant functions increases.

Correct
Correct:

8. How does the MDL principle handle the trade-off between model complexity and goodness of fit?
- ☐ By focusing solely on maximizing the goodness of fit.
- ☐ By always choosing the simplest possible model.
- ☐ By disregarding model complexity in favor of data fit.
- ☒ By penalizing more complex models unless they significantly improve data fit.

Correct
Correct: MDL penalizes complexity and favors models that fit well without being overly complex.

9. In supervised learning, what kind of data is used to train models?
- ☐ High-dimensional data
- ☐ Experimental data
- ☒ Labeled data
- ☐ Unlabeled data

Correct

10. In k-fold cross-validation, what does 'k' represent?
- ☒ The number of groups that the data is split into for the validation process.
- ☐ The proportion of data used for testing.
- ☐ The number of parameters in the model.
- ☐ The maximum number of iterations allowed in the validation process.

Correct
Correct: 'k' refers to the number of subsets the data is divided into for cross-validation.

11. What is the main purpose of adding a regularization term to a machine learning model's loss function?
- ☐ To automatically select the most relevant features in the dataset.
- ☒ To reduce overfitting by penalizing more complex models.
- ☐ To increase the model's prediction accuracy on the training data.
- ☐ To enhance the model's speed and computational efficiency.

Correct
Correct: Regularization aims to reduce overfitting by penalizing model complexity.

12. Which of the following best describes the relationship between Bayesian inference and Bootstrapping?
- ☒ Bayesian inference is a technique for estimating the uncertainty of model parameters, while Bootstrapping is a technique for quantifying the uncertainty of an estimator.
- ☐ Bayesian inference and Bootstrapping are both techniques used for parameter estimation.
- ☐ Bayesian inference and Bootstrapping have no connection and are used for entirely different purposes.
- ☐ Bootstrapping is a technique used to generate more data, while Bayesian inference is a technique for dealing with pre-existing data.

Correct
Correct: Both Bayesian inference and Bootstrapping are techniques for quantifying uncertainty, but they approach the problem from different angles.

13. Which of the following Python code snippets correctly represents the implementation of a basic Bayesian inference procedure?
- ☒ `prior = [1/3, 1/3, 1/3] likelihood = [0.3, 0.6, 0.1] posterior = [p * l / sum(p * l for p, l in zip(prior, likelihood))] for p, l in zip(prior, likelihood):`
- ☐ `prior = np.array([0.3, 0.6, 0.1]) likelihood = np.array([0.3, 0.6, 0.1]) posterior = prior * likelihood`
- ☐ `prior = np.array([0.3, 0.6, 0.1]) likelihood = np.array([0.3, 0.6, 0.1]) posterior = prior * likelihood / np.sum(prior * likelihood)`
- ☐ `prior = [1/3, 1/3, 1/3] likelihood = [0.3, 0.6, 0.1] posterior = [p * l for p, l in zip(prior, likelihood)]`

Incorrect
This is incorrect. You need to use numpy arrays for efficient, element-wise operations, not Python lists.

14. Which of the following is true about time series data?
- ☒ The presence of autocorrelation
- ☐ The lack of temporal order
- ☐ It is always continuous and numerical
- ☐ The number of data categories

Correct

15. In maximum likelihood inference, what is the consequence of choosing a model that is too complex for the data?
- ☐ The likelihood will be minimized
- ☒ The model will underfit the data
- ☐ The likelihood will be maximized
- ☐ The model will overfit the data

Incorrect
A complex model is more likely to overfit the data, not underfit.